Paper summary 8

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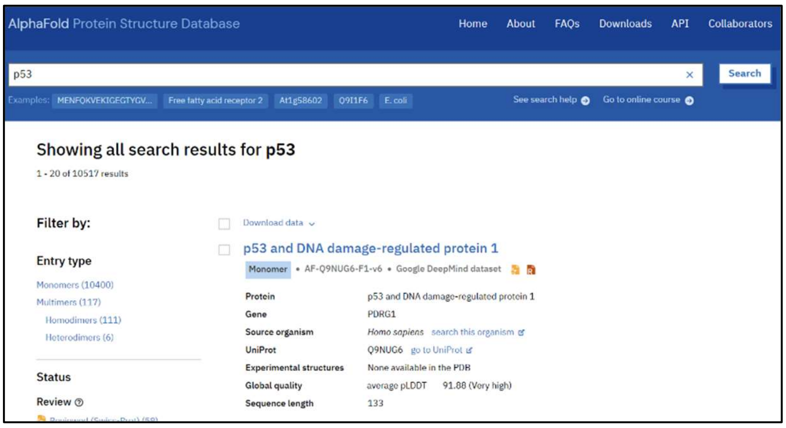
Introduction

According to the introduction of AlphaFold on the website, AlphaFold is a platform that uses artificial intelligence (Deep Learning) to predict the three-dimensional structure of proteins from amino acid sequences with high accuracy. In the past, a lot of time and cost were needed to determine protein structures, but AlphaFold has accelerated biological research and drug discovery.

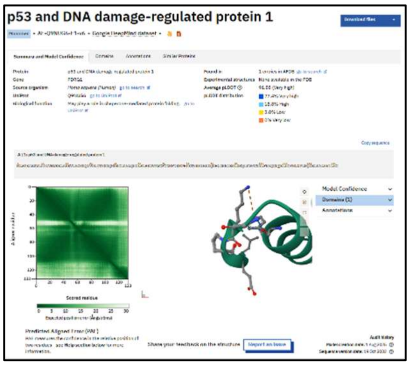
After AlphaGo defeated Lee Sedol in a Go match, people started to see the possibility of using artificial intelligence for protein structure prediction. AlphaFold's performance started from CASP13 and continued in CASP14, and the AlphaFold paper was published in Nature. DeepMind collaborated with EMBL-EBI to release the AlphaFold Protein Structure Database (AFDB), and researchers became able to use this data for biological research and drug discovery studies. In 2022, the predicted structures of almost all proteins were released, and AlphaFold 3 was announced through collaboration with Isomorphic Labs. Through this, it became possible to predict not only protein structures but also interactions between biomolecules such as DNA, RNA, and ligands. In addition, AlphaFold Server was released, allowing researchers to perform structure prediction through a web environment. AlphaFold has grown into a global research platform, and researchers are using it to carry out various studies.

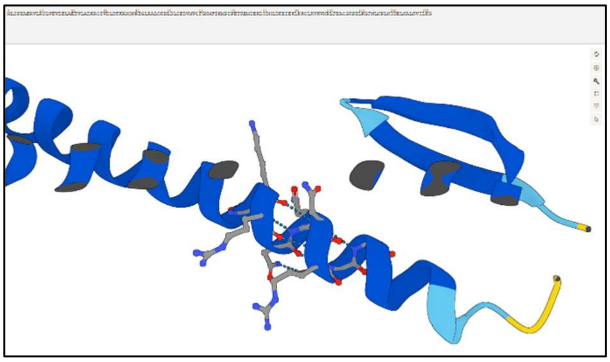
Database

The AlphaFold Database is freely available and contains more than 200 million protein structure data to accelerate prediction research. Users can search for proteins using a protein name, gene name, or UniProt ID, and can visualize protein structures in 3D and download them. In addition, it is possible to check the confidence score (pLDDT), which represents the prediction confidence. Users can also download large-scale protein databases and integrate AlphaFold data into pre-built pipelines using APIs.



The figure above shows the result page after searching for the p53 protein. Data related to the p53 protein, as well as data containing proteins related to p53, are provided. Information such as Protein, Gene, Source organism, UniProt, Experimental structures, Global quality, and Sequence length is provided. Through the filter function, it was possible to select the entry type or specify organism information, making the results easier to view.

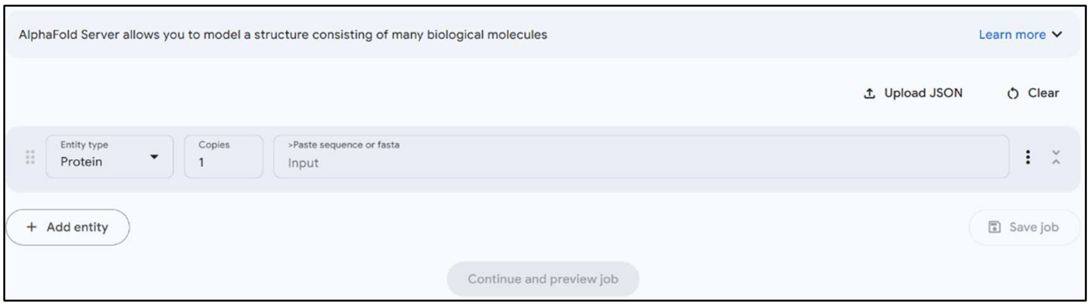




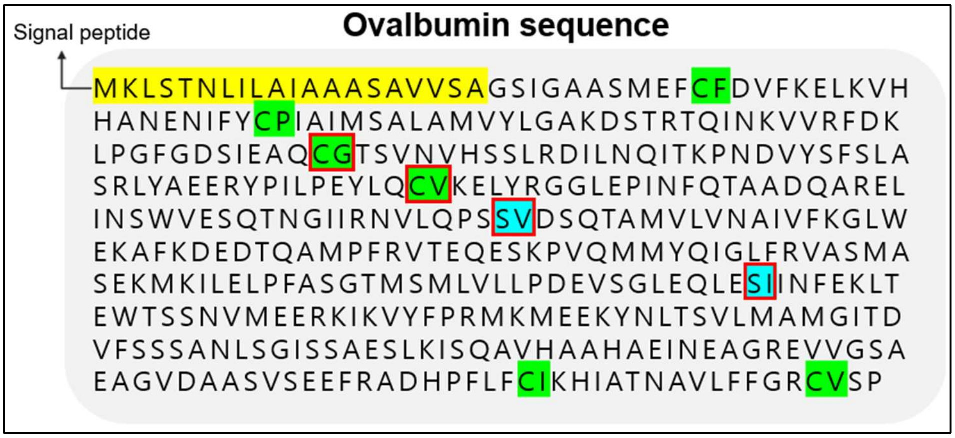
When I clicked on a searched protein structure to see more detailed information, I could view the amino acid sequence and the regions corresponding to each amino acid by rotating the structure with the mouse. In addition, the surrounding structure and binding information of the amino acids were displayed as figures. I was also able to check the Predicted Aligned Error (PAE) Matrix through a heatmap, which shows how accurately the position of residue j was predicted when aligned based on residue i. I clicked and rotated the structure with the mouse, and it felt like playing with LEGO blocks when I was young.

Server

While the Database allows users to explore previously known protein information, the Server seemed to be a place where users could examine protein structures used in their own research. AlphaFold Server is a web service based on the latest AlphaFold 3 model, and it can predict the structures and interactions of various biomolecules, including proteins, DNA, RNA, ligands, and ions, with high accuracy. It also supports structure prediction considering chemical modifications of proteins and nucleic acids, making it useful for life science research and drug discovery. In the input window shown in the figure below, users can specify the Entity type and copies, and enter a sequence to obtain protein structure information.

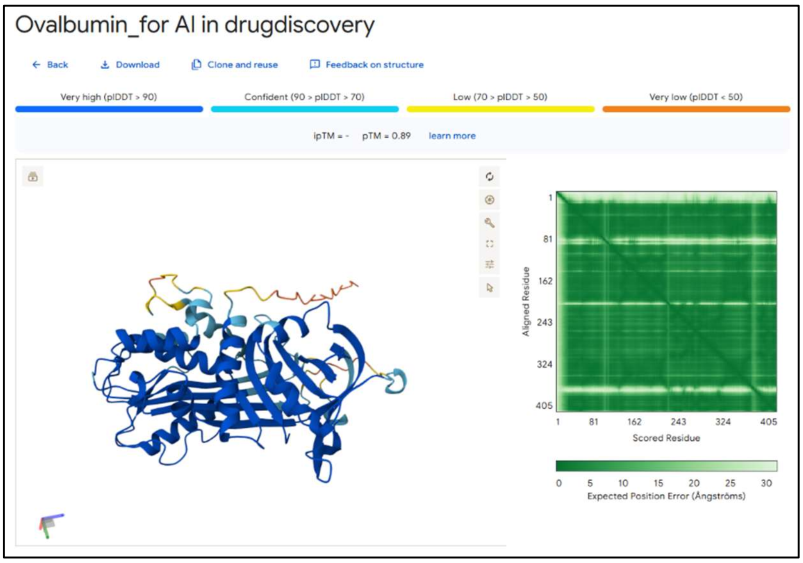


Coincidentally, my graduation research involved producing ovalbumin protein by cleaving an amino acid sequence, and I decided to enter the amino acid sequence that I had studied into the server. The figure below shows the ovalbumin sequence that was used in my graduation research.



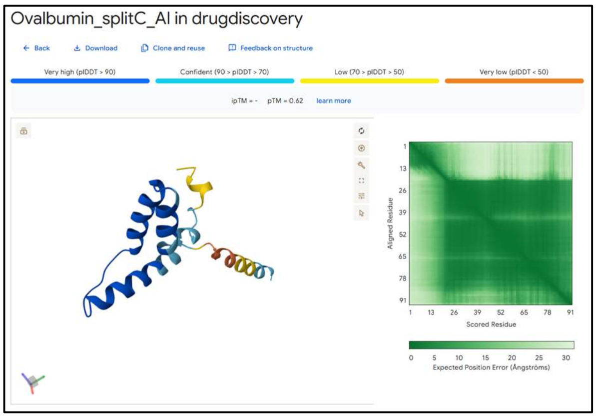
This amino acid sequence was used in a synthetic biology study in which a protein that was difficult to produce in yeast due to its large size was divided into two parts based on the region marked with a red box in the figure, and then combined in the medium through a trans-splicing process. After entering the sequence, it was displayed on the interface in groups of ten amino acids, making it very easy to read.

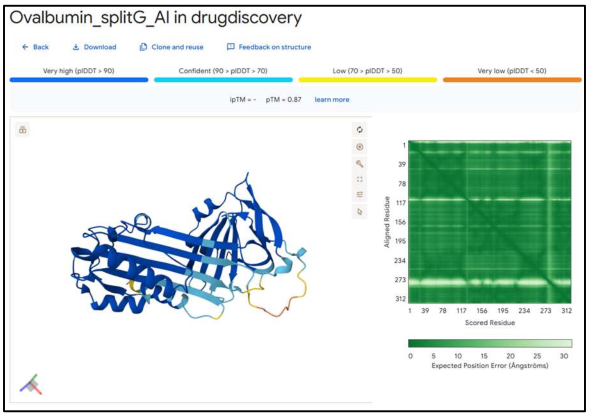
I was able to examine the structure of the ovalbumin protein that was used in my research. At the top of the figure, information about pLDDT, ipTM, and pTM was displayed. pLDDT (predicted Local Distance Difference Test) is a score



that indicates how confidently AlphaFold predicts the position of each amino acid. ipTM (interface predicted TM-score) is a score that evaluates the accuracy of binding interfaces between protein-protein, protein-DNA, and protein-RNA in complex predictions. pTM (predicted TM-score) is a value that evaluates the accuracy of the overall three-dimensional structure of a protein. Considering these values, I judged that the predicted structure of ovalbumin showed very high confidence overall. Most amino acids showed high pLDDT values, and the pTM value was 0.89, indicating that the overall structure was predicted with high accuracy. In addition, since the PAE matrix appeared dark overall, it could be judged that the structure was predicted properly.

Next, I entered the amino acid sequences that were divided based on the first red box (CG), which had been selected as a cleavage site in my previous research. In my research, I had predicted that protein expression would be the lowest when the sequence was cleaved at this position, but it actually showed the highest protein expression level.





By examining the structures of the cleaved ovalbumin proteins in this way, I thought that AlphaFold could be used to evaluate in advance which cleavage positions are more likely to maintain stable structures when producing ovalbumin as two independent protein fragments. I also thought that it might be possible to investigate whether the reason for the difference between the predicted expression level and the actual result was related to structural factors.

In the previous study, all four cleavage candidates were tested by Western blotting to confirm whether each protein fragment was successfully produced. I thought that if AlphaFold were used, it might be possible to examine the structures and filter out cleavage sites that do not need to be tested experimentally before performing the actual experiments. Of course, my experimental interpretation is still limited, and I do not know much about this field, so I think there may be some logical leaps in my reasoning.

Conclusion

Through this exploration, I learned several concepts and tools related to protein structure analysis. I learned that AlphaFold provides confidence metrics such as pLDDT, pTM, ipTM, and PAE Matrix, which help users evaluate the reliability of predicted protein structures. I also learned that the AlphaFold Database is a valuable data resource containing millions of protein structures, while AlphaFold Server allows researchers to predict structures from their own sequences. These tools can be useful for understanding protein structures and for selecting candidates before conducting experimental studies.

While exploring the AlphaFold Database and Server, I found it very interesting because it was much more sophisticated than I had expected from only reading about it. In my undergraduate graduation research, I failed to correctly predict the expression level of ovalbumin protein, but because experimental research requires a lot of time, I finished the project without conducting further studies. Looking at this program, I thought that I might have been able to analyze the reason more quickly and at a lower cost. I think that the development of AlphaFold will greatly accelerate research based on protein structure prediction, and this will make significant contributions to biological research and drug discovery research.